Appendix C: Fish Screening Project Descriptions

Diversions were evaluated for fish screening prioritization if they were within 400ft of expected anadromous distribution. This buffer distance allowed for any small location inaccuracies in the underlying data. Criteria used for prioritization included size of diversion (relative to stream size), number of fish species affected (anadromous and resident native species), and impact to fish. This third criteria, impact to fish, was determined using best professional judgment from ODFW fish biologists familiar with the diversions. More information on criteria can be found in Section 2.6.1 of the main K3RP report. For a full detailed description of methods, see Appendix E. The prioritization produced a list of fish screening projects within 'High,' 'Medium,' and 'Low,' priority tiers, and each project is described below.

Additional maps and KMZ files of the following projects are available for download on the K3RP website, https://k3rp-psmfc.hub.arcgis.com/

Screening Project List

Project Maps

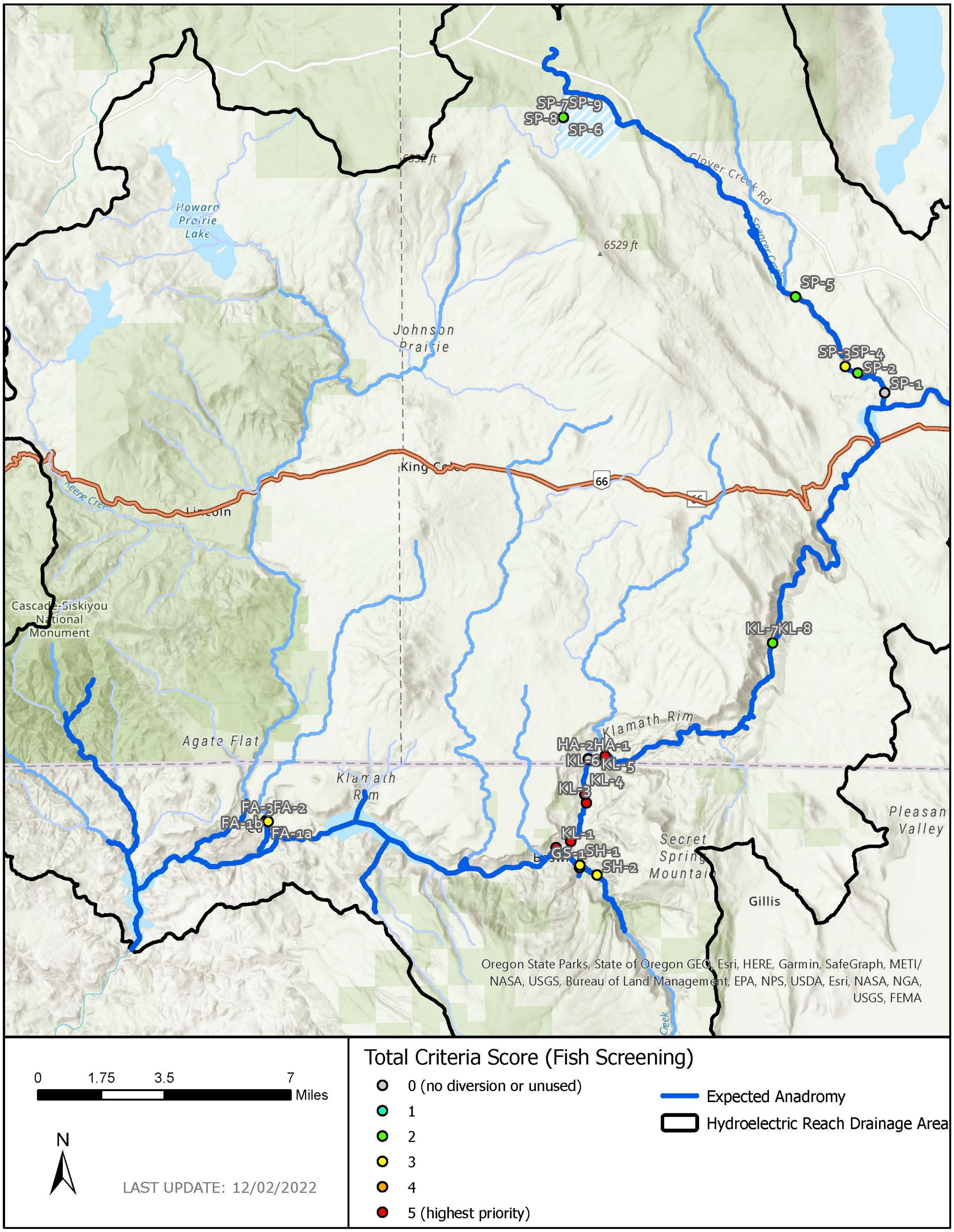

Figure 1. Map of fish screening projects downstream of Keno Dam.

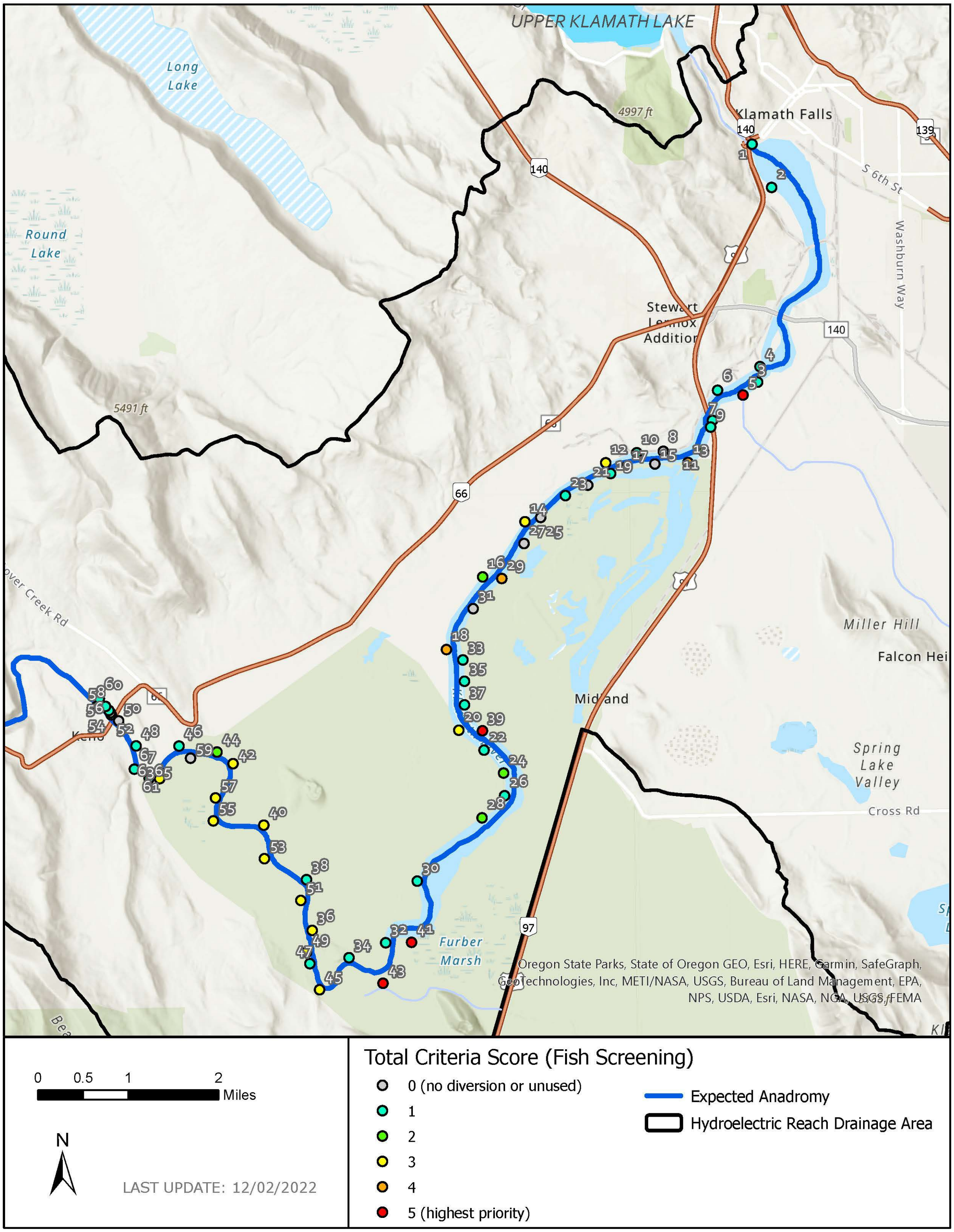

Figure 2. Map of fish screening projects upstream of Keno Dam ('Keno Reach'). Label names have been shortened for map (e.g. KENO-35 is 35). Projects on right bank are even numbers and projects on left bank are odd numbers.

Project# FA-1a

Figure 3. Diversion at FA-1a. City of Yreka 'Dam B.' June 2021.

Source: Fall Creek **State:** California

Approximate Total Volume: 15 cfs (total from FA-1a and FA-1b)

Screen: No

Size score: 5 **Benefit score**: 5 **Fish Impact:** 3 **Total score**: 3.8 **Priority Tier:** High

Coordinates: FA-1a: 41.985684, -122.362987

River Bank:

Notes: City of Yreka 'Dam B.' Secondary diversion for Yreka water supply. Used in conjunction with FA-1a. This diversion is planned to be blocked by a fish passage barrier installed through agreements defined in the Amended Klamath Hydro Settlement Agreement and screening will likely no longer be necessary.

Water Right Details:
City of Yreka, Application ID A022551

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# FA-1b

Figure 4. Diversion at FA-1b. City of Yreka 'Dam A.' June 2021.

Source: Unnamed tributary to Fall Creek (created by powerhouse tailwater) **State:** California

Approximate Total Volume: 15 cfs (total from FA-1a and FA-1b)

Screen: No

Size score: 5 **Benefit score**: 5 **Fish Impact:** 1 **Total score**: 2.6 **Priority Tier:** Medium

Coordinates: 41.985364, -122.361768

River Bank:

Notes: City of Yreka 'Dam A.' Primary diversion for Yreka water supply. Used in conjunction with FA-1b. This diversion is planned to be blocked by a fish passage barrier installed through agreements defined in the Amended Klamath Hydro Settlement Agreement and screening will likely no longer be necessary.

Water Right Details:
City of Yreka, Application ID A022551

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# FA-2

(No Photo)

Source: Unnamed tributary to Fall Creek (created by powerhouse tailwater) **State:** California

Approximate Total Volume: 10 cfs

Screen: No

Size score: 5 **Benefit score**: 5 **Fish Impact:** 1 **Total score**: 2.6 **Priority Tier:** Medium

Coordinates: 41.985281, -122.361956 (shares POD with FA-3)

River Bank:

Notes: Used in winter, structure preservation, reported use is 2cfs, non-consumptive. Overlap with FA-3. Construction of new Fall Creek hatchery may change usage and any project should be closely coordinated with CDFW. This diversion is planned to be blocked by a fish passage barrier installed through agreements defined in the Amended Klamath Hydro Settlement Agreement and screening will likely no longer be necessary.

Water Right Details:
PacifiCorp, Application ID S012966

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# FA-3

(No Photo)

Source: Unnamed tributary to Fall Creek (created by powerhouse tailwater) **State:** California

Approximate Total Volume: 10 cfs

Screen: No

Size score: 5 **Benefit score**: 5 **Fish Impact:** 1 **Total score**: 2.6 **Priority Tier:** Medium

Coordinates: 41.985282, -122.361956 (shares POD with FA-2)

River Bank:

Notes: Structure preservation to keep Fall Creek Hatchery running, reported use is 2cfs, non-consumptive. Overlap with FA-2. Construction of new Fall Creek hatchery may change usage and any project should be closely coordinated with CDFW. This diversion is planned to be blocked by a fish passage barrier installed through agreements defined in the Amended Klamath Hydro Settlement Agreement and screening will likely no longer be necessary.

Water Right Details:
CA Dept of Fish and Wildlife, Application ID A025896

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# SH-1

Figure 5. Fish screen at SH-1. June 2021.

Source: Shovel Creek **State:** California

Approximate Total Volume: 8 cfs

Screen: Yes

Size score: 5 **Benefit score**: 5 **Fish Impact:** 1 **Total score**: 2.6 **Priority Tier:** Medium

Coordinates: 41.966744, -122.194159

River Bank: Left

Notes: Irrigate 80 acres, used annually

Water Right Details:
PacifiCorp, Application ID S015379

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# SH-2

Figure 6. Fish screen at SH-2. June 2021.

Source: Shovel Creek **State:** California

Approximate Total Volume: 2 cfs

Screen: Yes

Size score: 4 **Benefit score**: 5 **Fish Impact:** 1 **Total score**: 2.4 **Priority Tier:** Low

Coordinates: 41.962759, -122.18499

River Bank: Right

Notes: Stockwater, 200head, used annually, measured (maximum reported measurement is 2-3cfs).

Water Right Details:
PacifiCorp, Application ID S015381

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# GS-1

Figure 7. Fish Screen at GS-1. June 2021.

Source: Grouse Springs Creek **State:** California

Approximate Total Volume: 5 cfs

Screen: Yes

Size score: 4 **Benefit score**: 5 **Fish Impact:** 1 **Total score**: 2.4 **Priority Tier:** Low

Coordinates: 41.965783, -122.19434

River Bank: Left

Notes: Stockwater for 250 head, used annually, measured (maximum reported measurement is 3cfs).

Water Right Details:
PacifiCorp, Application ID S015380

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# HA-1

(Identified in desktop analysis but could not find during survey)

(No Photo)

Source: Hayden Creek **State:** Oregon

Approximate Total Volume: 0.001 cfs

Screen: no diversion

Priority Tier: None

Coordinates: 42.009223, -122.189026

River Bank: n/a

Notes: could not find

Water Right Details:
BLM Claim: KA 373 * HL

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# HA-2

(Identified in desktop analysis but could not find during survey)

(No Photo)

Source: Hayden Creek **State:** Oregon

Approximate Total Volume: 0.1 cfs

Screen: no diversion

Priority Tier: None

Coordinates: 42.009443, -122.189063

River Bank: n/a

Notes: could not find

Water Right Details:
PacifiCorp Claim: KA 220 * IR

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# SP-1

(Identified in desktop analysis but could not find during survey)

(No Photo)

Source: Spencer Creek **State:** Oregon

Approximate Total Volume: 3 cfs

Screen: no diversion

Priority Tier: None

Coordinates: 42.155196, -122.027214

River Bank: n/a

Notes: could not find

Water Right Details:
ODFW Cert: 8118 OR * FI

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# SP-2

(No Photo)

Source: Spencer Creek **State:** Oregon

Approximate Total Volume: 0.1 cfs

Screen: unknown

Size score: 1 **Benefit score**: 5 **Fish Impact:** 1 **Total score**: 1.8 **Priority Tier:** Low

Coordinates: 42.16322, -122.041667

River Bank: unknown

Notes: Irrigation

Water Right Details:
Claim:KA 176 * IR Simmers

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# SP-3

Figure 8. Diversion location for SP-3 and SP-4. July 2021.

Source: Spencer Creek **State:** Oregon

Approximate Total Volume: 0.24 cfs

Screen: No

Size score: 1 **Benefit score**: 5 **Fish Impact:** 2.5 **Total score**: 2.7 **Priority Tier:** Medium

Coordinates: 42.165924, -122.048365 (shares POD with SP-4)

River Bank: Right

Notes: Irrigation

Water Right Details:
Claim: KA 174 * IR Anderson

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# SP-4

Photo: see Figure 8

Source: Spencer Creek **State:** Oregon

Approximate Total Volume: 0.24 cfs

Screen: No

Size score: 1 **Benefit score**: 5 **Fish Impact:** 2.5 **Total score**: 2.7 **Priority Tier:** Medium

Coordinates: 42.165924, -122.048365 (shares POD with SP-3)

River Bank: Right

Notes: Irrigation

Water Right Details:
Claim: KA 175 * IR James

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# SP-5

(No Photo)

Source: Spencer Creek **State:** Oregon

Approximate Total Volume: 0.15 cfs

Screen: No

Size score: 1 **Benefit score**: 5 **Fish Impact:** 1 **Total score**: 1.8 **Priority Tier:** Low

Coordinates: 42.194111, -122.074783

River Bank: unknown

Notes: Industrial/Manufacturing uses

Water Right Details:
Cert: 48027 OR * IM Weyerhaeuser Co.

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# SP-6

(No Photo)

Source: Unnamed Tributary to Spencer Creek **State:** Oregon

Approximate Total Volume: 9 cfs (may overlap with SP-7, SP-8, and SP-9)

Screen: unknown

Size score: 5 **Benefit score**: 5 **Fish Impact:** 1 **Total score**: 2.6 **Priority Tier:** Medium

Coordinates: 42.26712, -122.19929

River Bank: unknown

Notes: Irrigation

Water Right Details:
Claim: KA 141 * IR Charley

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# SP-7

(No Photo)

Source: Tunnel Creek (tributary to Spencer Creek) **State:** Oregon

Approximate Total Volume: 9 cfs (may overlap with SP-6, SP-8, and SP-9)

Screen: unknown

Size score: 5 **Benefit score**: 5 **Fish Impact:** 1 **Total score**: 2.6 **Priority Tier:** Medium

Coordinates: 42.267052, -122.199311

River Bank: unknown

Notes: Irrigation

Water Right Details:
Claim: KA 141 * IR Charley

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# SP-8

(No Photo)

Source: Unnamed Spring (tributary to Spencer Creek) **State:** Oregon

Approximate Total Volume: 18 cfs (may overlap with SP-6, SP-7, and SP-9)

Screen: unknown

Size score: 5 **Benefit score**: 5 **Fish Impact:** 1 **Total score**: 2.6 **Priority Tier:** Medium

Coordinates: 42.267048, -122.199296

River Bank: unknown

Notes: Irrigation

Water Right Details:
Claim: KA 141 * IR Charley

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# SP-9

(No Photo)

Source: Unnamed Spring (tributary to Spencer Creek) **State:** Oregon

Approximate Total Volume: 0.3 cfs (may overlap with SP-6, SP-7, and SP-8)

Screen: unknown

Size score: 2 **Benefit score**: 5 **Fish Impact:** 1 **Total score**: 2 **Priority Tier:** Low

Coordinates: 42.267133, -122.199352

River Bank: unknown

Notes: Irrigation

Water Right Details:
Claim: KA 141 * IR Charley

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KL-1

(No Photo)

Source: Klamath River **State:** California

Approximate Total Volume: paper right unknown, maximum reported diversion rate is 15-16cfs

Screen: No

Size score: 3 **Benefit score**: 5 **Fish Impact:** 5 **Total score**: 4.6 **Priority Tier:** High

Coordinates: 41.973856, -122.206866

River Bank: Right

Notes: 'Beswick Diversion.' Crosses Edge Creek. Irrigation of ~60acres, stockwater for 200 head, used annually. Reported measured use is maximum of 15-16cfs (reported rate was used for size scoring).

Water Right Details:
PacifiCorp, Application ID S015736

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KL-2

(No Photo)

Source: Klamath River **State:** California

Approximate Total Volume: 5 cfs

Screen: No

Size score: 2 **Benefit score**: 5 **Fish Impact:** 5 **Total score**: 4.4 **Priority Tier:** High

Coordinates: 41.976339, -122.198968

River Bank: Right

Notes: 'Owens Island Diversion.' Irrigation of ~8 acres, stockwater, 100head, used annually. Reported measured use is maximum of 4-5cfs (reported rate was used for size scoring).

Water Right Details:
PacifiCorp, Application ID S015378

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KL-3

(No Photo)

Source: Klamath River **State:** California

Approximate Total Volume: paper right unknown, max measured diversion rate is 8-9cfs

Screen: No

Size score: 2 **Benefit score**: 5 **Fish Impact:** 5 **Total score**: 4.4 **Priority Tier:** High

Coordinates: 41.991719, -122.190371

River Bank: Left

Notes: 'Home Field Diversions.' Irrigation of ~60 acres, stockwater, 200head, used annually. Reported measured use is maximum of 8-9cfs (reported rate was used for size scoring).

Water Right Details:
PacifiCorp, Application ID S015737

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KL-4

(No Photo)

Source: Klamath River **State:** California

Approximate Total Volume: 10 cfs

Screen: No

Size score: 2 **Benefit score**: 5 **Fish Impact:** 5 **Total score**: 4.4 **Priority Tier:** High

Coordinates: 41.995148, -122.191117

River Bank: Right

Notes: 'Owens Ditch.' Irrigation of ~43 acres, stockwater, 200head, used annually. Reported measured use is maximum of 5-6cfs (reported rate was used for size scoring).

Water Right Details:
PacifiCorp, Application ID S015377

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KL-5

Figure 9. KL-5 diversion (red arrow) and mainstem Klamath River flow (blue arrow). September 2019. Photo credit: ODFW.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 1.82 cfs

Screen: No

Size score: 1 **Benefit score**: 5 **Fish Impact:** 5 **Total score**: 4.2 **Priority Tier:** High

Coordinates: 42.008351, -122.184777

River Bank: Right

Notes: Irrigation of ~54 acres. Crosses Hayden Creek. NOAA, ODFW, and HSU visited the diversion in 2019 and provided the following information.

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

There is an unscreened diversion on river right of the Klamath River in Oregon that is designed to capture a portion of the Klamath River's flow using an angled boulder weir and ditch system (Figure 9, Figure 10). This ditch is 7 feet wide and parallels the river, intersects with Hayden Creek, and continues for 1700 feet before irrigating property belonging to PacifiCorp in CA. Hayden Creek drains directly into the ditch and we found water leaking out of a small breach in the ditch at a natural depression, likely from the natural channel of Hayden Creek. It appears that the majority of Hayden Creek water was going into the diversion and escaping at a blown out, tarped section of the ditch (Figure 12). It is possible that when the tarp is not in place (when irrigation water is not needed) all the water from the diversion is going back into the river at this point creating attraction flow in which adult fish then migrate up Hayden Creek to spawn. This theory warrants further investigation in the late winter/early spring.

The ditch is in poor shape and has several areas of failure where water is leaking out toward the Klamath River (Figure 11, Figure 12). The ditch is surrounded by reed canary grass (RCG), but it is not growing in the main ditch channel probably due to the depth of the ditch (2-4 ft) and the velocities that prohibit channel RCG growth.

We took flow measurements, temperature and dissolved oxygen measurements in the ditch at the point of diversion at 10am when the Klamath River was reading 2260 cfs at the Keno USGS gage. The diversion was taking 21.5 cfs at the point of diversion, the temperature was 18.2 C, and the dissolved oxygen (DO) was 6.15 mg/l. We electrofished several spots in the ditch within 50 ft of Hayden Creek and saw several juvenile O. Mykiss, dace, and signal crayfish entrained in the ditch (Figure 13).

Water Right Details:
PacifiCorp, KA 222

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

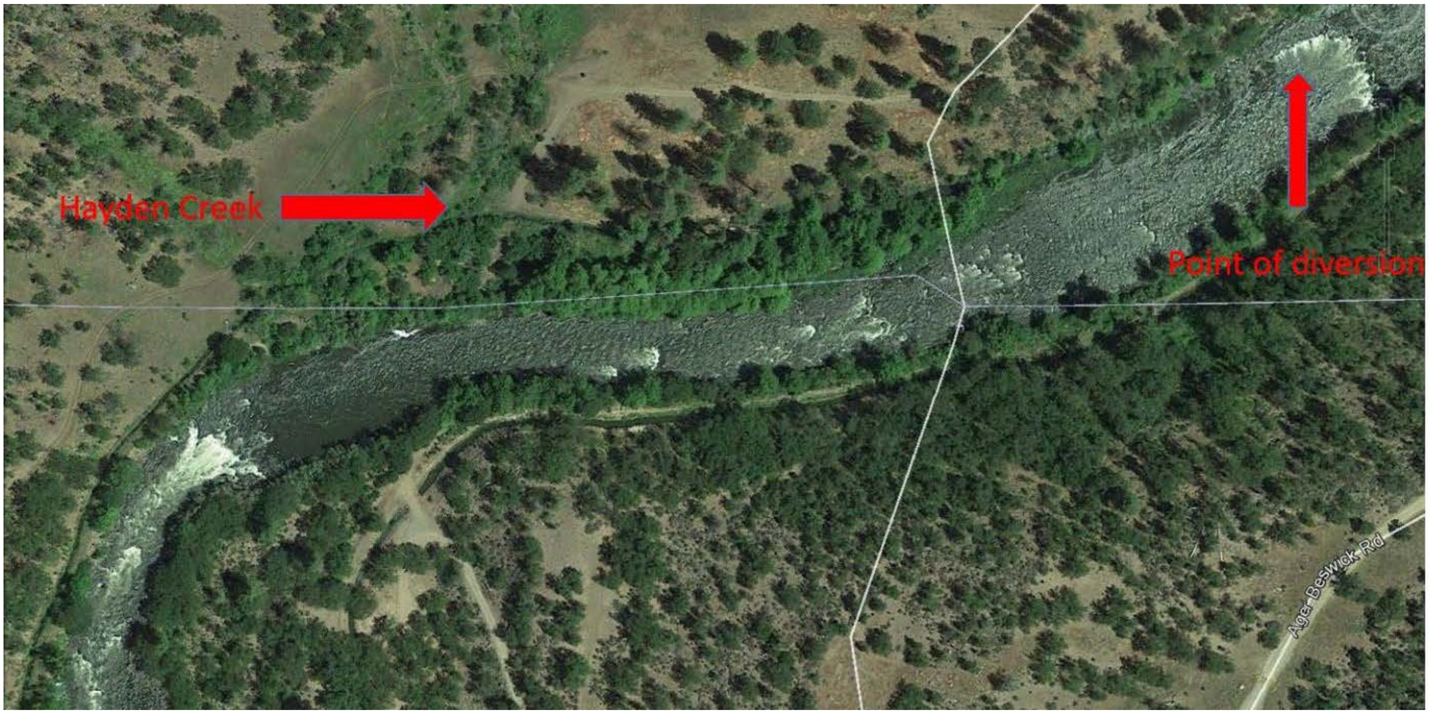

Figure 10. Map showing the point of diversion on the Klamath River and where it crosses Hayden Creek.

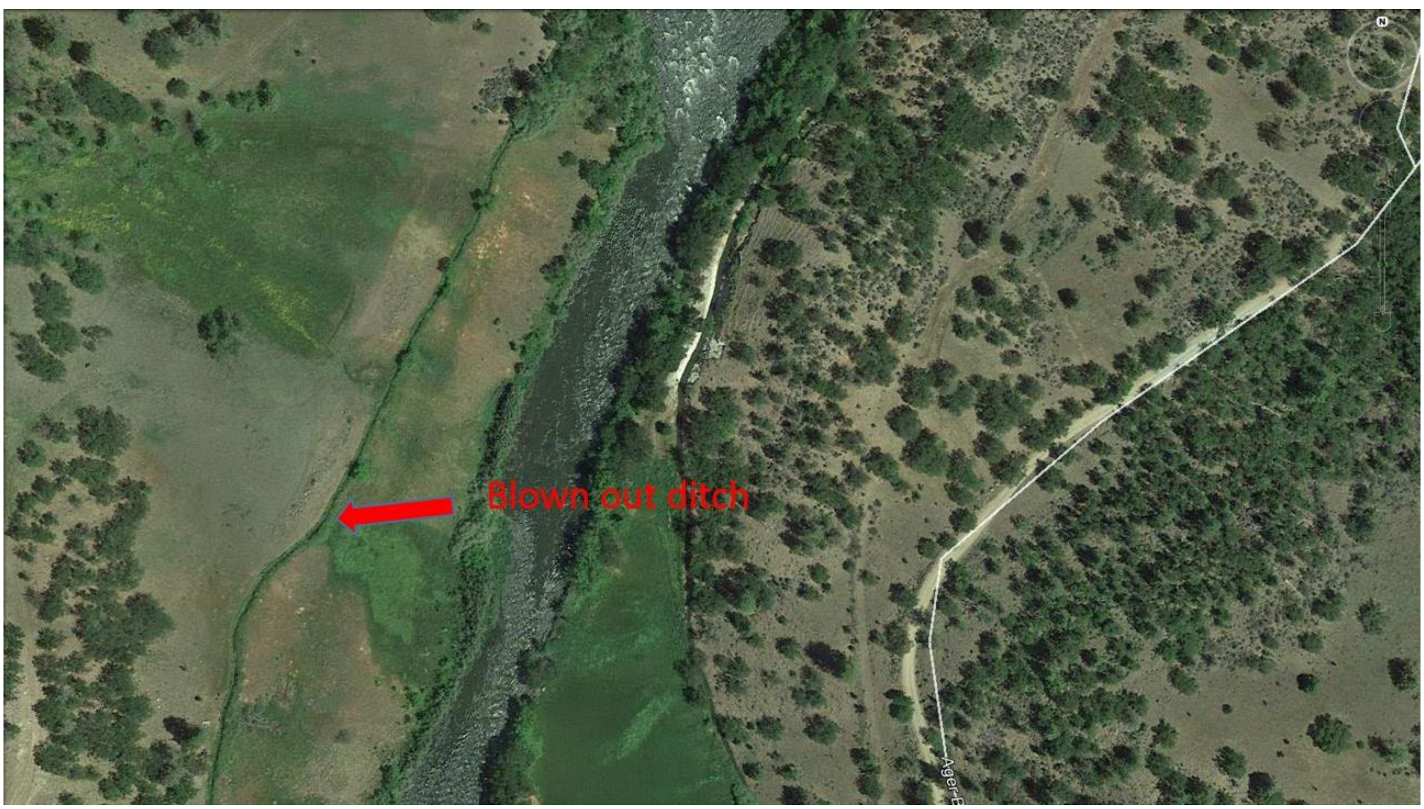

Figure 11. Map showing leaky ditch prior to reaching the point of use downstream

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Figure 12. Picture of ditch damage and tarps used to fix the leak approximately 50 ft west of where the ditch crosses Hayden Creek. Salmonids and Dace were found in the channel created by this failure. Photo credit: ODFW.

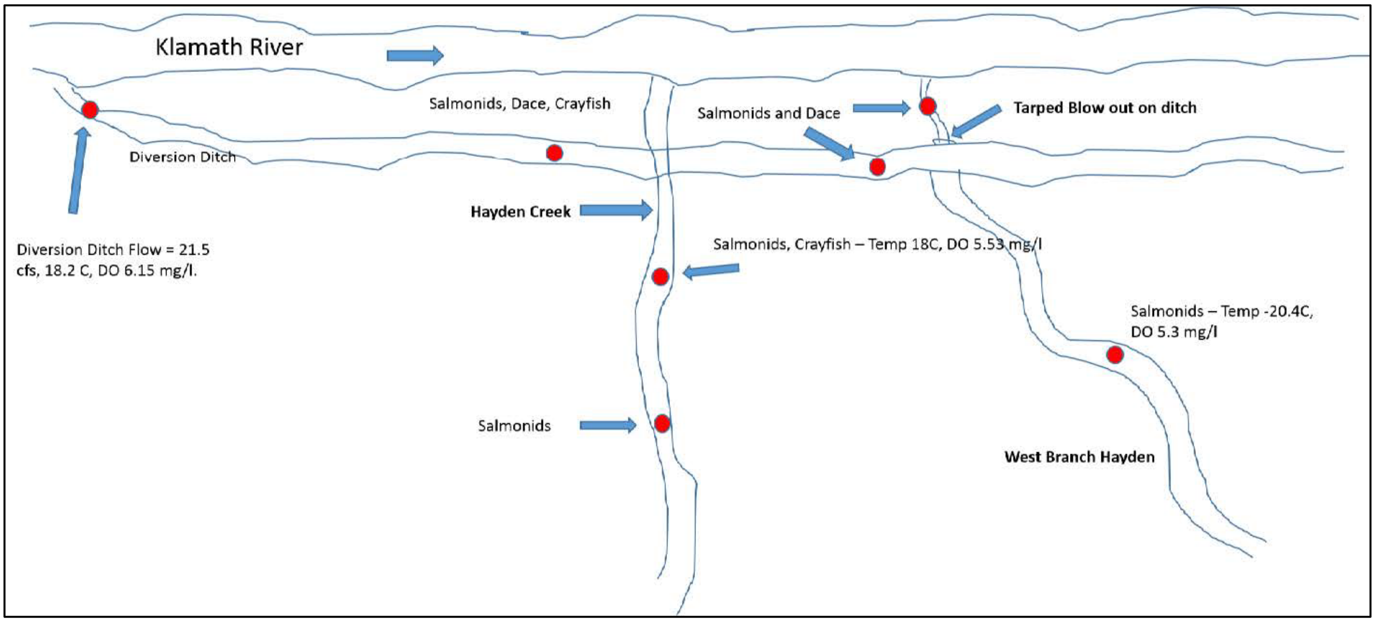

Figure 13. Illustration of Lower Hayden Creek site conditions (not to scale), sampling locations and flow/temp/DO/fish presence. Map credit: ODFW

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KL-6

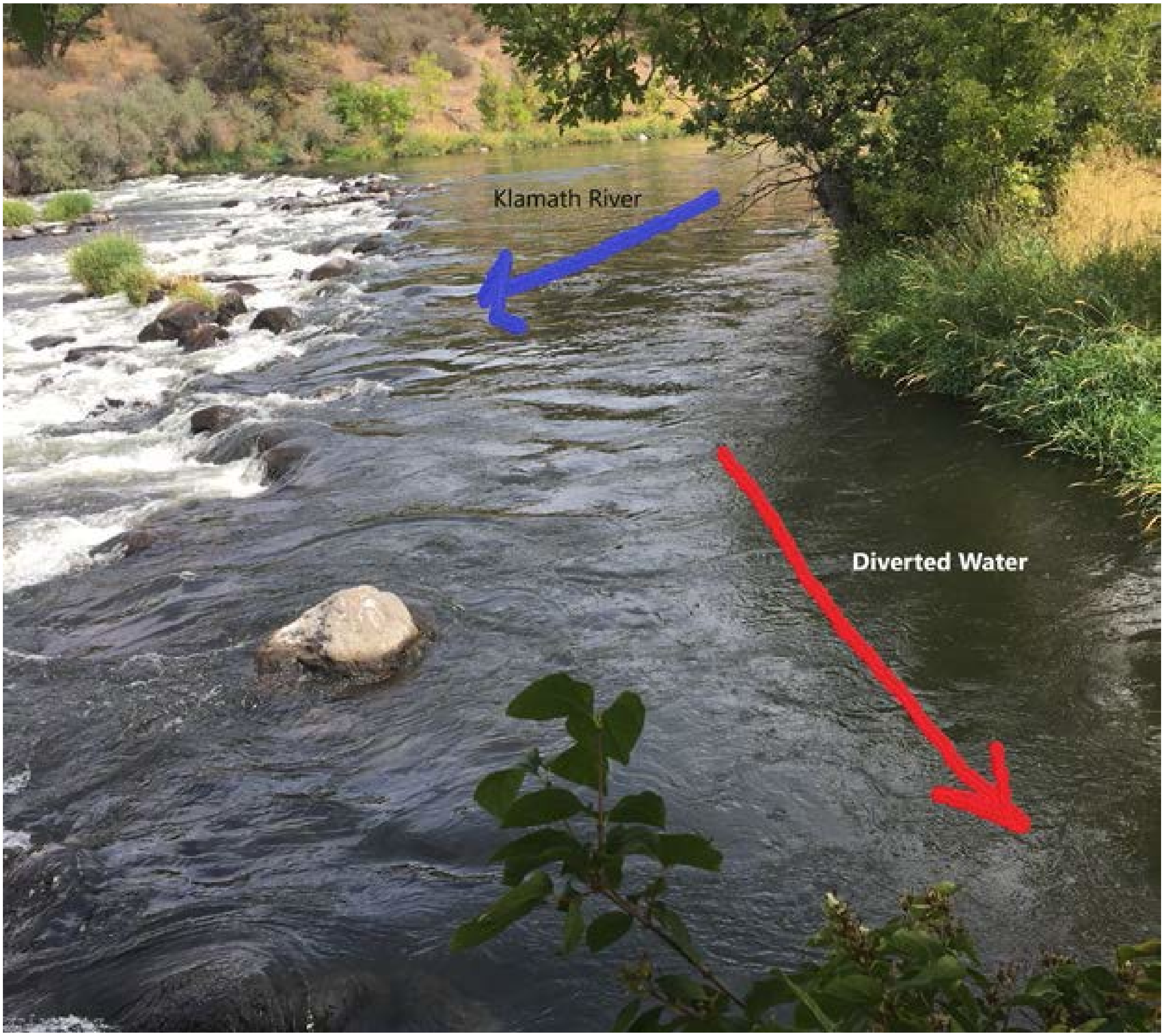

Figure 14. KL-6 diversion (red arrow) and mainstem Klamath River flow (blue arrow). September 2018. Photo credit: ODFW. September 2018.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 1.51 cfs

Screen: No

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Size score: 1 **Benefit score**: 5 **Fish Impact:** 5 **Total score**: 4.2 **Priority Tier:** High

Coordinates: 42.010474, -122.179978

River Bank: Left

Notes: Irrigation of ~30 acres. ODFW visited the diversion in 2018 and provided the following information:

Diversion was accessed via Topsy Grade Rd. from Keno. The POD is located on BLM land. The POD cannot be reached by vehicle, road down to POD is cabled off and road is highly eroded away. Large Boulders are placed across the river to guide water into an excavated channel leading to the head gate. Boulders across the channel are probably more influential in channeling water when river is not peaking from JC Boyle Powerhouse. Flows in the Klamath River were around 1700 cfs (peaking) when the site was visited. Rough estimate of amount of water diverted was around 30-45 cfs (Figure 15). Head gate is minimally maintained, as vegetation is overgrown around the gate. The diversion ditch can be accessed about a half mile downstream where it appears to be maintained via an ATV trail that is seldom used. 0.65 miles downstream is a river raft take out (BLM Stateline Takeout) and campground. Damage to the canal is prevalent at this point, where water is leaving the ditch and running down the road to the takeout (Figure 16).

Water Right Details:
PacifiCorp, KA 221

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Figure 15. Culvert behind headgate on KL-6 in 2018. ODFW estimated flow of ~30 cfs. Photo credit: ODFW.

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Figure 16. Water from KL-6 diversion flowing over dirt road in 2018. Photo credit: ODFW.

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KL-7

(No Photo)

Source: Klamath River **State:** Oregon

Approximate Total Volume: 0.63 cfs

Screen: Unknown

Size score: 1 **Benefit score**: 5 **Fish Impact:** 1 **Total score**: 1.8 **Priority Tier:** Low

Coordinates: 42.055213, -122.089164

River Bank: Left

Notes: Irrigation

Water Right Details:
Cert: 739 OR * IR

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

<u>Project# KL-8</u>

(No Photo)

Source: Klamath River **State:** Oregon

Approximate Total Volume: 0.56 cfs

Screen: Unknown

Size score: 1 **Benefit score**: 5 **Fish Impact:** 1 **Total score**: 1.8 **Priority Tier:** Low

Coordinates: 42.055213, -122.089164

River Bank: Left

Notes: Agricultural Uses

Water Right Details:
Cert: 739 OR * AG

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-1

(No Photo)

Source: Klamath River **State:** Oregon

Approximate Total Volume: 0.11 cfs

Screen: Unknown

Size score: 1 **Benefit score**: 4 **Total score**: 2.5 **Priority Tier:** Low

Coordinates: 42.219078, -121.788944

River Bank: Left (in Lake Euwana)

Notes: Did not visually assess

Water Right Details:
CERT 67078 0.11 CFS

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-2

Figure 17. Headgate at KENO-2, October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 1.55 cfs

Screen: No

Size score: 1 **Benefit score**: 4 **Total score**: 2.5 **Priority Tier:** Low

Coordinates: 42.212093, -121.784852

River Bank: Right (in Lake Euwana)

Notes: 18" headgate (approx.), was regulated off by OWRD in October 2021 when photo was taken.

Water Right Details:
KA 1000 1.55 CFS Modoc Lumber

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-3

Figure 18. Diversion KENO-3. Arrow points to headgate at end of ditch. October 2021

Source: Klamath River **State:** Oregon

Approximate Total Volume: 2.34 cfs

Screen: No

Size score: 1 **Benefit score**: 4 **Total score**: 2.5 **Priority Tier:** Low

Coordinates: 42.180854, -121.788473

River Bank: Left

Notes: 18" headgate (approx.) set back about 50' from river at end of canal

Water Right Details:
KA 194 2.34 CFS Griffith IR

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-4

Figure 19. KENO-4 location. Diversion is presumably located in red pumphouse. October 2021

Source: Klamath River **State:** Oregon

Approximate Total Volume: 0.18 cfs

Screen: No

Size score: 1 **Benefit score**: 4 **Total score**: 2.5 **Priority Tier:** Low

Coordinates: 42.183344, -121.787911

River Bank: Right

Notes: pump in old red brick pumphouse

Water Right Details:
CERT 67512 0.18 CFS

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-5, Lost River Diversion Ditch

Figure 20. KENO-5, Lost River Diversion Ditch. Photo shows photo shows the entire width of the very wide diversion canal. The Klamath River flows to the right out of the photo. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 1903 cfs

Screen: No

Size score: 5 **Benefit score**: 4 **Total score**: 4.5 **Priority Tier:** High

Coordinates: 42.178817, -121.791674

River Bank: Left

Notes: Lost River Diversion Ditch, open canal at river, radial gates on canal 0.65 miles from river, can flow both directions, used year round. During normal water years and in the spring, this canal flows water from the Lost River to the Klamath River; however, in drier years and during the summer, this canal flows water from the Klamath River to the Lost River.

Water Right Details:
Water rights included in the total volume are listed below but may be inaccurate, due to the complexity of the water rights within the Klamath Project.

KA 312 383.6 CFS USFWS
KA 318 100.4 CFS USFWS
KA 316 0.6 CFS USFWS
KA 315 9.0 CFS USFWS
KA 313 284.4 CFS USFWS
KA 319 10.0 CFS USFWS

KA 1000 0.5 CFS Griffith HG
KA 1000 650 CFS KID/TID
KA 1000 100 CFS KID/TID drain1
KA 1000 105 CFS Miller Hill PP
KA 1000 KID PP 1-10 total 10 CFS
KA 317 249.5 CFS USFWS

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-6

Figure 21. Timber operation at location of KENO-6. October 2021

Source: Klamath River **State:** Oregon

Approximate Total Volume: 1.33

Screen: Unknown

Size score: 1 **Benefit score**: 4 **Total score**: 2.5 **Priority Tier:** Low

Coordinates: 42.179637, -121.797154

River Bank: Right

Notes: Unclear if/how diversion works. Located in middle of timber operation in log pond and can discharge to Klamath River

Water Right Details:
KA 1000 REAMES 1.33 CFS

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-7

Figure 22. Diversion at KENO-7. October 2021

Source: Klamath River **State:** Oregon

Approximate Total Volume: 1.77 cfs

Screen: Unknown

Size score: 1 **Benefit score**: 4 **Total score**: 2.5 **Priority Tier:** Low

Coordinates: 42.174808, -121.798343

River Bank: Left

Notes: Open canal at river

Water Right Details:
CERT 65444 1.77 CFS

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-8

(Identified in desktop analysis but could not find during survey)

Source: Klamath River **State:** Oregon

Approximate Total Volume: 10.81 cfs

Screen: No diversion found

Priority Tier: None

Coordinates: 42.169962, -121.809055

River Bank: Right

Notes: could not find, could be combined with a nearby diversion

Water Right Details:
CERT 77011 2.68 CFS Holliday IR
CERT 5589 7.13 CFS Plevna IR
CERT 21504 1 CFS IM

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-9

Figure 23. Diversion at KENO-9. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 1.02 cfs

Screen: No

Size score: 1 **Benefit score**: 4 **Total score**: 2.5 **Priority Tier:** Low

Coordinates: 42.173743, -121.798747

River Bank: Left

Notes: Open canal at river

Water Right Details:
KA 193 1.02 CFS Moates IR

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-10

Figure 24. Timber operation at location of KENO-10. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 1.124 cfs

Screen: Unknown

Size score: 1 **Benefit score**: 4 **Total score**: 2.5 **Priority Tier:** Low

Coordinates: 42.169779, -121.814807

River Bank: Right

Notes: Unclear if/how the diversion works. Located in the middle of wood processing operation and can discharge to river

Water Right Details:
Cert 48887 0.674 CFS (IM)
Cert 48886 0.45 CFS (IM)

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-11

Figure 25. Diversion at KENO-11. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 9 cfs

Screen: No

Size score: 2 **Benefit score**: 4 **Total score**: 3 **Priority Tier:** Medium

Coordinates: 42.169815, -121.800349

River Bank: Left

Notes: Open canal at river

Water Right Details:
KA 1000 9 CFS Plevna

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-12

Figure 26. Diversion at KENO-12. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 20.4 cfs

Screen: No

Size score: 3 **Benefit score**: 4 **Total score**: 3.5 **Priority Tier:** High

Coordinates: 42.168242, -121.821525

River Bank: Right

Notes: Pump, industrial intake with debris screen

Water Right Details:
CERT 37577 20.4 IM, AC

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-13

Figure 27. Diversion and Fish Screen at KENO-13. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 31.15 cfs

Screen: Yes

Size score: 4 **Benefit score**: 4 **Total score**: 4 **Priority Tier:** High (but already screened)

Coordinates: 42.168109, -121.803774

River Bank: Left

Notes: 2 options: pump or headgate. Pump has conveyor belt screen, headgate is unscreened. When pump doesn't work (at certain river levels), headgate is used (according to ODFW). Miller Island #1

Water Right Details:
KA 186 0.03 CFS ODFW
CERT 28569 2.12 CFS ODFW
KA 1000 POD #1 Total CFS 29.0 CFS Miller Island

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-14

Figure 28. Diversion at KENO-14. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 14.81 cfs

Screen: Unknown

Size score: 3 **Benefit score**: 4 **Total score**: 3.5 **Priority Tier:** High

Coordinates: 42.158957, -121.839141

River Bank: Right

Notes: Unclear if/how diversion functions. Heavily vegetated open canal at river.

Water Right Details:
KA 1000 14.0 CFS Plevna #1 & #2
CERT 5130 0.81 CFS Talbot IR

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-15

(Identified in desktop analysis but during survey found that diversion is unused)

Figure 29. Unused diversion at KENO-15. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 1.56 cfs

Screen: no, diversion not used

Priority Tier: None

Coordinates: 42.167939, -121.810917

River Bank: Left

Notes: 24" headgate, no screen, not used (according to ODFW), corresponds to Miller Island #2

Water Right Details:
KA 186 1.56 CFS ODFW
KA 1000 Miller Island POD #2

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-16

Figure 30. Diversion at KENO-16. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 8 cfs

Screen: No

Size score: 2 **Benefit score**: 4 **Total score**: 3 **Priority Tier:** Medium

Coordinates: 42.150181, -121.848426

River Bank: Right

Notes: Open canal at river

Water Right Details:

KA 1000 8.0 CFS Plevna #3

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-17

(Identified in desktop analysis but during survey found that diversion is unused)

Figure 31. Unused diversion at KENO-17. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 0 cfs

Screen: no, diversion not used

Priority Tier: None

Coordinates: 42.167586, -121.816949

River Bank: Left

Notes: 2' slide gate, no screen, not used (according to ODFW), corresponds to Miller Island #3

Water Right Details:

KA 1000 Miller Island POD #3

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-18

Figure 32. Diversion at KENO-18. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 56.64 cfs

Screen: No

Size score: 4 **Benefit score**: 4 **Total score**: 4 **Priority Tier:** High

Coordinates: 42.138595, -121.856459

River Bank: Right

Notes: Open canal at river

Water Right Details:

KA 178 6.66 CFS Dobson IR
KA 185 5.88 CFS Keno ID
KA 185 6.81 CFS Keno ID
KA 1000 12.68 CFS PDIC
KA 1000 4.0 CFS Kerns
Cert 67576 4.66 CFS Emmitt IR
Cert 39228 / KA 178 3.98 CFS Murdoch (IR/DO)

Cert 29862 2.0 CFS Kerns IR
Cert 29532 1.0 CFS Hoppe IR
Cert 8264 1.19 CFS Kerns IR
Cert 4870 0.47 CFS Simmers IR
Cert 3772 2.5 CFS Kerns IR
Cert 37583 4.47 CFS (IR)
Cert 67638 0.34 CFS Johnston (Supplement)

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-19

Figure 33. Diversion at KENO-19. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 2.7 cfs

Screen: Yes

Size score: 1 **Benefit score**: 4 **Total score**: 2.5 **Priority Tier:** Low

Coordinates: 42.166509, -121.820494

River Bank: Left

Notes: 18" headgate, screened 0.5 miles down ditch (ODFW), Miller Island #4

Water Right Details:
KA 186 2.7 CFS ODFW
KA 1000 Miller Island POD #4

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-20

Figure 34. Diversion at KENO-20. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 10.48 cfs

Screen: No

Size score: 3 **Benefit score**: 4 **Total score**: 3.5 **Priority Tier:** High

Coordinates: 42.12561, -121.85402

River Bank: Right

Notes: Open culvert (could be drain), headgate just downstream of open culvert (could be diversion), but unclear exactly how it works.

Water Right Details:
KA 185 10.48 CFS Keno IR

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-21

(Identified in desktop analysis but during survey found that diversion is unused)

Figure 35. Unused diversion at KENO-21. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 1.38 cfs

Screen: no, diversion not used

Priority Tier: None

Coordinates: 42.164685, -121.825471

River Bank: Left

Notes: Open canal at river leads to pump close by, unscreened, not used (according to ODFW), corresponds to Miller Island #5

Water Right Details:
KA 186 1.38 CFS ODFW
KA 1000 Miller Island POD #5

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-22

Figure 36. Pumped diversion at KENO-22. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 1.25 cfs

Screen: Unknown

Size score: 1 **Benefit score**: 4 **Total score**: 2.5 **Priority Tier:** Low

Coordinates: 42.122359, -121.848638

River Bank: Right

Notes: Pump, likely unscreened but pipe intake was not visible

Water Right Details:
KA 185 1.25 CFS Keno ID IR

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-23

Figure 37. Screen and diversion at KENO-23. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 3.05

Screen: Yes

Size score: 1 **Benefit score**: 4 **Total score**: 2.5 **Priority Tier:** Low

Coordinates: 42.16306, -121.830332

River Bank: Left

Notes: Vertical panel screen on canal at river, primary ODFW diversion

Water Right Details:
KA 186 1.57 CFS ODFW
CERT 37582 0.52 FURBER IR IS
CERT 2804 0.96 CFS FERBER IR

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-24

Figure 38. Diversion at KENO-24. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 5.46 cfs

Screen: Unknown

Size score: 2 **Benefit score**: 4 **Total score**: 3 **Priority Tier:** Medium

Coordinates: 42.118636, -121.844491

River Bank: Right

Notes: 24" headgate, no screen at river, pump visible at end of canal approximately 500' from river

Water Right Details:

KA 185 5.46 CFS Keno ID IR

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-25

(Identified in desktop analysis but could not find during survey)

Figure 39. Unused diversion at KENO-25. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 5.46 cfs

Screen: no diversion

Priority Tier: None

Coordinates: 42.159603, -121.835732

River Bank: Left

Notes: could not find anything that looked like a diversion except for some wire panels behind a log, not used (according to ODFW), corresponds to Miller Island #6

Water Right Details:
KA 186 1.97 CFS ODFW
KA 1000 Miller Island POD #6

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-26

Figure 40. Diversion at KENO-26. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 4.41 cfs

Screen: No

Size score: 1 **Benefit score**: 4 **Total score**: 2.5 **Priority Tier:** Low

Coordinates: 42.114994, -121.844338

River Bank: Right

Notes: 24" headgate

Water Right Details:
KA 185 4.41 CFS Keno ID IR

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-27

(Identified in desktop analysis but could not find during survey)

(no photo)

Source: Klamath River **State:** Oregon

Approximate Total Volume: 4.41 cfs

Screen: no diversion

Priority Tier: None

Coordinates: 42.155454, -121.839407

River Bank: Left

Notes: Could not find anything that looked like a diversion except for an old power pole, not used (according to ODFW), corresponds to Miller Island #7

Water Right Details:
KA 186 1.01 CFS ODFW
KA 1000 Miller Island POD #7

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-28

Figure 41. Diversion at KENO-28. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 5.77 cfs

Screen: No

Size score: 2 **Benefit score**: 4 **Total score**: 3 **Priority Tier:** Medium

Coordinates: 42.111489, -121.849268

River Bank: Right

Notes: 24" headgate

Water Right Details:
KA 185 5.77 CFS Keno ID IR

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-29

Figure 42. Diversion at KENO-29. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 32.43 cfs

Screen: No

Size score: 4 **Benefit score**: 4 **Total score**: 4 **Priority Tier:** High

Coordinates: 42.149883, -121.844303

River Bank: Left

Notes: 24" headgate

Water Right Details:
KA 186 7.28 CFS ODFW
KA 183 1.26 CFS Vidricksen IR LV
CERT 6316 2.38 CFS Hooper IR
CERT 76030 2.14 CFS VanValken
CERT 8728 12.5 CFS Hooper IR
CERT 37581 2.4 CFS Furber IR, WI
CERT 37583 4.47 CFS Furber IR

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-30

Figure 43. Diversion at KENO-30. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: Unknown

Screen: No

Size score: 1 **Benefit score**: 4 **Total score**: 2.5 **Priority Tier:** Low

Coordinates: 42.101463, -121.863424

River Bank: Right

Notes: 24" headgate, new plastic culvert, unknown which water rights are associated with diversion. Diversion was seen during field survey but not identified in desktop analysis.

Water Right Details:
Unknown

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-31

(Identified in desktop analysis but could not find during survey)

(no photo)

Source: Klamath River **State:** Oregon

Approximate Total Volume: 0 cfs

Screen: no, diversion not used

Priority Tier: None

Coordinates: 42.145103, -121.850579

River Bank: Left

Notes: Could not find anything that looked like a diversion, not used (according to ODFW)

Water Right Details:
CER 5069 1.75 CFS VanValken

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-32

Figure 44. Screen and diversion at KENO-32. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 4.71 cfs

Screen: Yes

Size score: 1 **Benefit score**: 4 **Total score**: 2.5 **Priority Tier:** Low

Coordinates: 42.091632, -121.870417

River Bank: Right

Notes: Vertical panel screen on canal at river

Water Right Details:
KA 185 3.89 CFS Keno ID IR
KA 185 0.82 CFS Keno ID IR

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-33

Figure 45. Diversion at KENO-33. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 3.67 cfs

Screen: No

Size score: 1 **Benefit score**: 4 **Total score**: 2.5 **Priority Tier:** Low

Coordinates: 42.136894, -121.852937

River Bank: Left

Notes: 24" headgate

Water Right Details:
KA 183 3.67 CFS Vidricksen IR LV

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-34

Figure 46. Diversion at KENO-34. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 2.43 cfs

Screen: No

Size score: 1 **Benefit score**: 4 **Total score**: 2.5 **Priority Tier:** Low

Coordinates: 42.089313, -121.878364

River Bank: Right

Notes: Pump

Water Right Details:
KA 185 2.43 CFS Keno ID IR

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-35

Figure 47. Diversion at KENO-35. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 4.11 cfs

Screen: No

Size score: 1 **Benefit score**: 4 **Total score**: 2.5 **Priority Tier:** Low

Coordinates: 42.133463, -121.852658

River Bank: Left

Notes: 24" headgate

Water Right Details:
KA 177 4.11 CFS Anderson IR LV

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-36

Figure 48. Screen and diversion at KENO-36. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 12.74 cfs

Screen: Yes

Size score: 3 **Benefit score**: 4 **Total score**: 3.5 **Priority Tier:** High (but already screened)

Coordinates: 42.093772, -121.886227

River Bank: Right

Notes: vertical panel screen on canal at river

Water Right Details:
KA 185 12.74 CFS Keno ID IR

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-37

Figure 49. Diversion at KENO-37. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 1.78 cfs

Screen: No

Size score: 1 **Benefit score**: 4 **Total score**: 2.5 **Priority Tier:** Low

Coordinates: 42.129669, -121.852722

River Bank: Left

Notes: 18" headgate

Water Right Details:
KA 182 1.78 CFS Furber IR LV

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-38

Figure 50. Diversion at KENO-38. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 1.28 cfs

Screen: No

Size score: 1 **Benefit score**: 4 **Total score**: 2.5 **Priority Tier:** Low

Coordinates: 42.101931, -121.887308

River Bank: Right

Notes: 24" headgate

Water Right Details:
KA 185 1.28 CFS Keno ID IR

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-39, North Canal

Figure 51. North Canal, KENO-39. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 710.1 cfs

Screen: No

Size score: 5 **Benefit score**: 4 **Total score**: 4.5 **Priority Tier:** High

Coordinates: 42.125473, -121.848921

River Bank: Left

Notes: North Canal, open canal at river, fence to keep boats out

Water Right Details: Water rights included in the total volume are listed below but may be inaccurate, due to the complexity of the water rights within the Klamath Project.

KA 177 0.78 CFS Anderson IR
KA 1002 2.93 CFS KDD
KA 182 2.98 CFS Furber IR LV
KA 210 1.0 CFS Tule Smoke WI
KA 1000 200 CFS KDD

Permit 48435 15 CFS Klamath Hill ID (IR)
Permit 43334 480. CFS KDD IR
Cert 37581 4.8 CFS Furber (IR)
Cert 56928 2.61 CFS Long

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-40

Figure 52. Upstream and downstream diversions at KENO-40. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 10.55 cfs

Screen: No

Size score: 3 **Benefit score**: 4 **Total score**: 3.5 **Priority Tier:** High

Coordinates: 42.110768, -121.896374

River Bank: Right

Notes: 2, 18" headgates, 200ft apart, unscreened. Upstream one could be a drain, downstream one could be a diversion.

Water Right Details:
KA 185 7.46 CFS Keno ID IR
KA 185 3.09 CFS Keno ID IR

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-41, Ady Canal

Figure 53. Ady Canal, KENO-41. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 1102.6 cfs

Screen: No

Size score: 5 **Benefit score**: 4 **Total score**: 4.5 **Priority Tier:** High

Coordinates: 42.091646, -121.864832

River Bank: Left

Notes: Ady Canal, open canal at river, canal size limits flow to 350cfs

Water Right Details: Water rights included in the total volume are listed below but may be inaccurate, due to the complexity of the water rights within the Klamath Project.
KA 312 383.6 CFS USFWS
KA 315 9 CFS USFWS
KA 316 0.6 CFS USFWS
KA 313 284.4 CFS USFWS
KA 314 25 CFS USFWS
KA 1000 400CFS KDD

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-42

Figure 54. Upstream and downstream diversions at KENO-42. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 14.72 cfs

Screen: No

Size score: 3 **Benefit score**: 4 **Total score**: 3.5 **Priority Tier:** High

Coordinates: 42.120716, -121.902836

River Bank: Right

Notes: 2, 24"" headgates and one pump house, all unscreened. Unclear which is diversion, and which is drain.

Water Right Details:
KA 185 12.47 CFS Keno ID IR
KA 185 2.25 CFS Keno ID IR

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-43, Klamath Straits Drain

Figure 55. Klamath Straits Drain, KENO-43. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 543.02 cfs

Screen: No

Size score: 5 **Benefit score**: 4 **Total score**: 4.5 **Priority Tier:** High

Coordinates: 42.08513, -121.871103

River Bank: Left

Notes: Klamath Straits, open canal at river, only drains and does not divert from Klamath River. Because of this it may not be a priority for screening. However, since there are large flows into the Klamath River from the canal, it could create an attraction flow that could be detrimental to fish.

Water is diverted and used for the Klamath Project from this canal, but it is recirculated water that has been captured by the system. The following water rights were gathered from a desktop analysis of the canal and may represent those recirculated water rights. Additional information should be gathered from local experts before project planning.

Water Right Details:

KA 205 10 CFS Tucker IR
KA 1000 2.9 CFS ADIC #2
KA 1000 3.7 CFS ADIC #3
KA 1000 14.51 CFS ADIC #4
KA 1000 14.5 CFS ADIC #5
KA 1000 14.5 CFS ADIC #6
Cert 83180 1.95 Flowers IR
T- 5225 2.5 CFS Flowers IR
Permit 43334 480.46 CFS KDD IR

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-44

Figure 56. Diversion at KENO-44. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 8.44 cfs

Screen: No

Size score: 2 **Benefit score**: 4 **Total score**: 3 **Priority Tier:** Medium

Coordinates: 42.122614, -121.906267

River Bank: Right

Notes: Open canal at river

Water Right Details:
KA 185 3.12 CFS Keno ID IR
KA 185 0.39 CFS Keno ID IR
KA 204 0.09 CFS Evensizer IR
KA 1000 4.84 CFS Johnson Intake

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-45

Figure 57. Diversion at KENO-45. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 15.18 cfs

Screen: No

Size score: 3 **Benefit score**: 4 **Total score**: 3.5 **Priority Tier:** High

Coordinates: 42.08423, -121.884802

River Bank: Left

Notes: 24" headgate

Water Right Details:
KA 205 0.67 CFS Tucker IR
KA 1000 14.51 CFS ADIC #1

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-46

Figure 58. Diversion at KENO-46. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 4.31 cfs

Screen: Unknown

Size score: 1 **Benefit score**: 4 **Total score**: 2.5 **Priority Tier:** Low

Coordinates: 42.123654, -121.91445

River Bank: Right

Notes: Open canal at river, no screen at river, pump visible at end of canal approximately 1000' from river

Water Right Details:
CERT 48565 4.31 CFS Puckett IR

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-47

Figure 59. Diversion at KENO-47. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 1.63 cfs

Screen: No

Size score: 1 **Benefit score**: 4 **Total score**: 2.5 **Priority Tier:** Low

Coordinates: 42.088434, -121.886826

River Bank: Left

Notes: Pump

Water Right Details:
KA 203 1.63 CFS Calmes IR

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-48

Figure 60. Diversion at KENO-48. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 4.31 cfs

Screen: No

Size score: 1 **Benefit score**: 4 **Total score**: 2.5 **Priority Tier:** Low

Coordinates: 42.123766, -121.923707

River Bank: Right

Notes: 2 pumps next to each other

Water Right Details:
CERT 48565 4.31 CFS Puckett

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-49

Figure 61. Diversion at KENO-49. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 14.07 cfs

Screen: No

Size score: 3 **Benefit score**: 4 **Total score**: 3.5 **Priority Tier:** High

Coordinates: 42.090233, -121.887509

River Bank: Left

Notes: Open canal at river

Water Right Details:
KA 203 0.75 CFS Calmes
KA 207 #6 3.67 CFS Kite IR
CERT 38456 #7 9.65 CFS Kite IR

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-50

(Identified in desktop analysis but could not find during survey)

Figure 62. No diversion at KENO-50. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 1.3 cfs

Screen: no diversion

Priority Tier: None

Coordinates: 42.127818, -121.927402

River Bank: Right

Notes: could not find anything that looked like a diversion except for some old boards, likely not used

Water Right Details:
CERT 20273 1.3 CFS Puckett IR

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-51

Figure 63. Diversion at KENO-51. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 13.33 cfs

Screen: No

Size score: 3 **Benefit score**: 4 **Total score**: 3.5 **Priority Tier:** High

Coordinates: 42.098606, -121.888592

River Bank: Left

Notes: 24" headgate

Water Right Details:
KA 207 #5 3.68 CFS Kite IR
CERT 38456 #6 9.65 CFS Kite IR

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-52

Figure 64. Diversion at KENO-52. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 0.16 cfs

Screen: No

Size score: 1 **Benefit score**: 4 **Total score**: 2.5 **Priority Tier:** Low

Coordinates: 42.128798, -121.928971

River Bank: Right

Notes: Small domestic pump

Water Right Details:
KA 181 0.13 CFS Boehme IR
KA 184 0.01 CFS Pappas IR
KA 189 0.02 CFS Puckett

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-53

Figure 65. Diversion at KENO-53. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 13.33 cfs

Screen: No

Size score: 3 **Benefit score**: 4 **Total score**: 3.5 **Priority Tier:** High

Coordinates: 42.105389, -121.896337

River Bank: Left

Notes: 24" headgate and pump

Water Right Details:
KA 207 #4 3.68 CFS Kite
CERT 38456 #5 9.65 CFS Kite IR

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-54

(Identified in desktop analysis but could not find during survey)

(no photo)

Source: Klamath River **State:** Oregon

Approximate Total Volume: 0.02 cfs

Screen: no diversion

Priority Tier: None

Coordinates: 42.129146, -121.929158

River Bank: Right

Notes: could not find

Water Right Details:
KA 179 0.02 CFS Kenyon IR

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-55

Figure 66. Diversion at KENO-55. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 12.18 cfs

Screen: No

Size score: 3 **Benefit score**: 4 **Total score**: 3.5 **Priority Tier:** High

Coordinates: 42.111569, -121.907248

River Bank: Left

Notes: 2, 24" headgates and one pump house, all unscreened. Unclear which is diversion, and which is drain.

Water Right Details:
KA 206 3.85 CFS Peacore IR LV
KA 208 3.58 CFS IR
CERT 30280 0.31 CFS Howard IR
CERT 38457 3.52 CFS Howard IR
CERT 44404 0.3 CFS Hopkins IR
CERT 44491 0.62 CFS Howard

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-56

Figure 67. Diversion at KENO-56. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 0.09 cfs

Screen: No

Size score: 1 **Benefit score**: 4 **Total score**: 2.5 **Priority Tier:** Low

Coordinates: 42.129515, -121.929498

River Bank: Right

Notes: Small domestic pump, 2" PVC

Water Right Details:
KA 180 0.02 CFS Estenson IR
KA 188 0.03 CFS Powell IR
KA 190 0.04 CFS Batzer IR

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-57

Figure 68. Diversion at KENO-57. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 11.38 cfs

Screen: No

Size score: 3 **Benefit score**: 4 **Total score**: 3.5 **Priority Tier:** High

Coordinates: 42.115249, -121.906714

River Bank: Left

Notes: 30" headgate

Water Right Details:
KA 207 #3 1.73 CFS Kite IR
CERT 38456 #4 9.65 CFS Kite IR

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-58

Figure 69. Diversion at KENO-58. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 0.03 cfs

Screen: No

Size score: 1 **Benefit score**: 4 **Total score**: 2.5 **Priority Tier:** Low

Coordinates: 42.130166, -121.9302

River Bank: Right

Notes: Small domestic pump

Water Right Details:
KA 192 0.03 CFS Wagner IR

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-59

(Identified in desktop analysis but could not find during survey)

(no photo)

Source: Klamath River **State:** Oregon

Approximate Total Volume: 9.65 cfs

Screen: no diversion

Priority Tier: None

Coordinates: 42.121682, -121.911996

River Bank: Left

Notes: could not find, could be combined with a nearby diversion

Water Right Details:
CERT 38456 #3 9.65 CFS Kite IR

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-60

Figure 70. Diversion at KENO-60. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 0.05 cfs

Screen: No

Size score: 1 **Benefit score**: 4 **Total score**: 2.5 **Priority Tier:** Low

Coordinates: 42.131459, -121.93151

River Bank: Right

Notes: Small domestic pump, 3" hose

Water Right Details:
CERT 48599 0.05 CFS Crosslin IR

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-61

Figure 71. Diversion at KENO-61. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 11.37 cfs

Screen: No

Size score: 3 **Benefit score**: 4 **Total score**: 3.5 **Priority Tier:** High

Coordinates: 42.119207, -121.917618

River Bank: Left

Notes: 24" headgate

Water Right Details:
KA 207 #2 1.72 CFS Kite IR
CERT 38456 #2 9.65 CFS Kite IR

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-63

Figure 72. Diversion at KENO-63. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 11.37 cfs

Screen: No

Size score: 3 **Benefit score**: 4 **Total score**: 3.5 **Priority Tier:** High

Coordinates: 42.118458, -121.918716

River Bank: Left

Notes: Open canal at river

Water Right Details:
KA 207 #1 1.72 CFS Kite IR
CERT 38456 #1 9.65 CFS Kite IR

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-65

Figure 73. Diversions at KENO-65. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 1.5 cfs

Screen: No

Size score: 1 **Benefit score**: 4 **Total score**: 2.5 **Priority Tier:** Low

Coordinates: 42.118519, -121.920977

River Bank: Left

Notes: 24" headgate and 2 pumps

Water Right Details:
KA 201 0.86 CFS Berg
CERT 60389 0.25 CFS Rambo IR
CERT 40545 0.39 CFS Berg IR

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-67

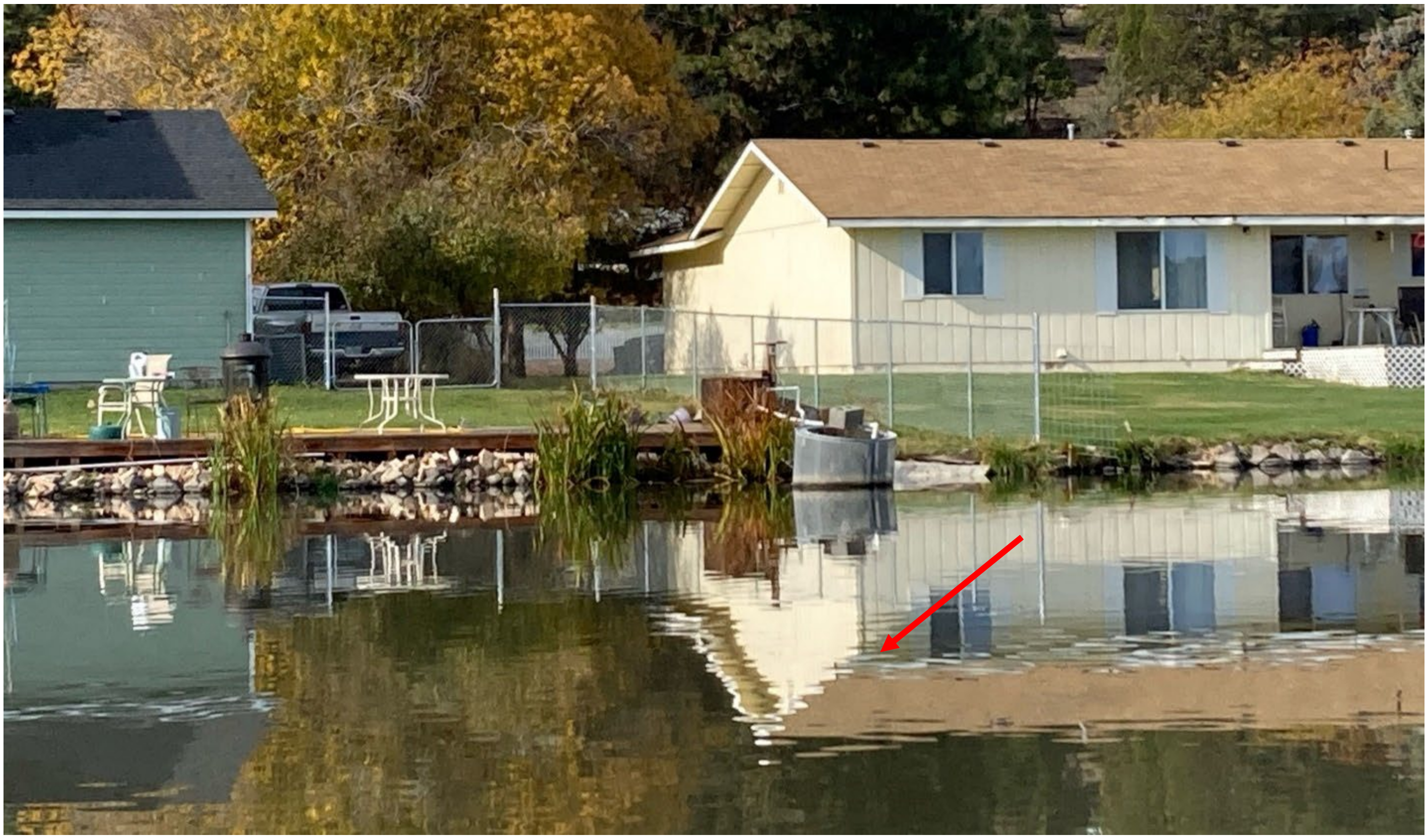

Figure 74. Diversion at KENO-67. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 0.03 cfs

Screen: No

Size score: 1 **Benefit score**: 4 **Total score**: 2.5 **Priority Tier:** Low

Coordinates: 42.120034, -121.924155

River Bank: Left

Notes: Small domestic pump, 2" PVC

Water Right Details:
CERT 48593 0.03 CFS Gray

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)

Project# KENO-69

Figure 75. Diversion at KENO-69. October 2021.

Source: Klamath River **State:** Oregon

Approximate Total Volume: 0.01 cfs

Screen: No

Size score: 1 **Benefit score**: 4 **Total score**: 2.5 **Priority Tier:** Low

Coordinates: 42.130214, -121.93206

River Bank: Left

Notes: Small domestic pump

Water Right Details:
CERT 65296 0.01 CFS Stewart

(Note that all water rights data is approximate and potentially inaccurate. Before undertaking any project scoping or design, it will be necessary to verify water rights, volumes, and usage.)